Network Layer: Control Plane Definitions (Ch. 5)

5.1 & 5.2 Introduction to Routing Algorithms

Control Plane The network-wide logic that determines the end-to-end paths packets take. Its primary output is the **forwarding table** used by the data plane in each router.
Routing Algorithm An algorithm that finds "good" (least-cost) paths from a source to a destination through a network of routers.
Graph Abstraction The modeling of a network as a graph G=(N,E), where N is the set of routers (nodes) and E is the set of links (edges). Each edge has a cost.
Link-State (LS) Algorithm A class of routing algorithm where every router has a complete map of the entire network topology and all link costs. Each router then independently computes its own forwarding table. This is a **global** information approach.
- **Dijkstra's Algorithm:** The classic LS algorithm used to compute the least-cost paths from a single source node to all other nodes in the graph.

Distance-Vector (DV) Algorithm A class of routing algorithm where a router only knows the cost to its direct neighbors. It iteratively exchanges path-cost estimates (its "distance vector") with its neighbors to converge on the least-cost paths. This is a **decentralized** information approach.
- **Bellman-Ford Equation:** The dynamic programming principle that underlies DV algorithms.
- **Count-to-Infinity Problem:** A slow convergence issue in DV algorithms where "bad news" (a link failure or cost increase) propagates slowly through the network, potentially causing routing loops.

Static vs. Dynamic Routing • **Static:** Routes are manually configured and do not change automatically.
- **Dynamic:** Routers automatically adapt to changes in link costs or topology.

5.3 & 5.4 Intra-AS and Inter-AS Routing

Autonomous System (AS) A group of routers under the same administrative control (e.g., an ISP network). The internet is a collection of interconnected ASes.
Intra-AS Routing (Interior Gateway Protocol - IGP) The routing protocol used *within* a single AS. The focus is on performance and finding the optimal path. All routers in the AS must run the same IGP.
- **OSPF (Open Shortest Path First):** The most common intra-AS protocol. It is a **link-state** protocol that uses Dijkstra's algorithm. OSPF can be structured into a **hierarchical** two-level design with a backbone area (Area 0) and local areas.

Inter-AS Routing (Exterior Gateway Protocol - EGP) The routing protocol used *between* different ASes. The focus is on enforcing policy.
- **BGP (Border Gateway Protocol):** The de facto inter-AS routing protocol of the internet. It is a **Path Vector** protocol.

Gateway Router A router at the "edge" of an AS that has a direct link to a router in another AS.

5.4 BGP (Border Gateway Protocol) in Detail

eBGP (external BGP) A BGP session between routers in *different* ASes. Used to advertise reachability to external networks.
iBGP (internal BGP) A BGP session between routers *within the same* AS. Used to propagate external reachability information to all internal routers of the AS.
Path Vector Protocol A protocol where route advertisements include the entire path of ASes the advertisement has traversed.
AS-PATH A key BGP attribute that lists the sequence of ASes a route advertisement has passed through. It is the primary mechanism for loop detection.
NEXT-HOP A BGP attribute that indicates the IP address of the router interface that begins the AS-PATH.
Policy-Based Routing The practice of choosing routes based on administrative and economic policies rather than just performance metrics. BGP's primary function is to enable policy. An AS can implement an *import policy* to filter which routes it accepts from its neighbors.
BGP Route Selection The process a router uses to select the best path when it learns multiple routes to the same destination. The hierarchy of rules is typically:
- Highest **Local Preference** value (policy decision).
- Shortest **AS-PATH**.
- Closest **NEXT-HOP** router (known as **Hot Potato Routing**).
- Additional criteria.

Hot Potato Routing The selfish routing strategy of choosing a path that gets the packet out of one's own AS as quickly as possible, minimizing the cost for the local network. This is achieved by selecting the route with the closest NEXT-HOP

router according to the intra-AS protocol (OSPF).

5.5 The SDN Control Plane

SDN (Software-Defined Networking) An architecture that separates the network's control plane from the data plane and logically centralizes it.
SDN Controller (Network OS) A logically centralized entity that acts as the "brain" of the network. It maintains a global view of the network state, provides an API for network-control applications, and installs forwarding rules into the data plane switches.
Northbound API The interface between the SDN controller and the network-control applications. It allows applications to specify their desired network behavior.
Southbound API The interface between the SDN controller and the data plane switches (e.g., OpenFlow). It allows the controller to program the forwarding tables of the switches.

5.6 & 5.7 ICMP and Network Management

ICMP (Internet Control Message Protocol) A network-layer protocol used by hosts and routers to communicate network-level information, primarily for **error reporting**. ICMP messages are carried within IP datagrams.
- **Destination Unreachable (Type 3):** An ICMP error message indicating a packet could not be forwarded.
- **Time Exceeded (Type 11):** An ICMP error message sent by a router when a packet's TTL reaches zero. This is the mechanism that makes 'traceroute' work.
- **Echo Request/Reply (Types 8/0):** The messages used by the 'ping' utility.

SNMP (Simple Network Management Protocol) A classic application-layer protocol for network management, using a manager-agent model to query and set variables on network devices.
- **MIB (Management Information Base):** A database of managed objects on a device that defines the structure of its configuration and state data.

YANG A data modeling language used to define the structure, meaning, and constraints of configuration and state data on a network device.
NETCONF A modern, transaction-based protocol (using YANG models) for configuring network devices. It is more robust than SNMP for complex configuration tasks.